

GUSTAVUS, AK, USA

WMO: 703670

Lat: 58.425N

Lon: 135.707W

Elev: 40

StdP: 14.67

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25322

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	6.8	-13.7	2.6	6.2	-8.0	3.6	13.7	24.6	39.9	20.9	39.0	1.6	350	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	71.1	57.3	66.2	56.0	63.7	55.1	59.5	66.5	58.0	64.2	56.8	62.2	7.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.3	67.5	60.9	55.0	64.4	59.9	54.2	62.6	59.2	25.9	66.6	25.0	63.8	24.2	62.0	65.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.5	14.8	-7.2	76.5	6.6	4.2	-11.9	79.5	-15.8	82.0	-19.6	84.3	-24.4	87.4		
(5)			-6.6	61.8	6.3	2.2	-11.2	63.4	-14.9	64.7	-18.4	66.0	-23.0	67.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	29.2	29.9	33.1	39.9	47.0	53.0	56.1	55.9	50.1	42.2	33.7	30.5
	DBStd	11.83	9.71	8.54	7.15	4.58	4.44	3.84	2.99	3.17	3.92	4.98	7.83	9.47
	HDD50	3539	646	563	525	303	114	9	0	1	43	243	488	604
	HDD65	8474	1111	983	990	752	558	360	277	284	446	706	938	1069
	CDD50	541	0	0	0	1	21	99	189	182	47	2	0	0
	CDD65	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH74	57	0	0	0	0	1	18	18	20	0	0	0	0
	CDH80	6	0	0	0	0	0	2	3	1	0	0	0	0
	WSAvg	5.0	4.6	5.1	6.1	5.5	5.1	4.3	3.9	4.0	4.8	5.7	5.6	5.8
	PrecAvg	71.4	7.4	5.8	5.2	3.6	3.0	2.4	3.3	4.6	8.5	10.2	8.7	8.7
(15) Precipitation	PrecMax	95.4	13.2	9.3	11.7	6.7	6.1	4.4	6.1	8.6	12.0	16.2	16.1	17.1
	PrecMin	58.0	3.3	2.0	0.8	1.0	0.7	0.9	1.0	1.7	3.0	4.7	2.8	4.5
	PrecStd	8.9	2.4	2.1	2.2	1.4	1.4	0.8	1.3	1.7	1.9	2.6	2.8	3.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	46.4	52.1	63.5	71.2	77.0	75.3	76.6	67.9	55.4	50.1	48.3
		MCWB	43.9	40.8	43.8	48.5	53.3	58.8	59.5	59.2	54.8	49.1	45.7	44.3
	2%	DB	44.7	45.0	46.4	56.7	64.3	69.5	67.6	70.5	63.1	53.7	46.4	45.5
		MCWB	40.9	40.2	40.4	45.0	50.9	56.1	58.2	58.4	52.6	48.4	42.8	42.1
	5%	DB	42.5	42.9	44.8	52.6	61.1	64.0	64.2	65.2	59.4	51.9	45.1	43.1
		MCWB	39.2	38.8	39.4	43.1	49.6	54.5	57.0	57.0	52.1	47.7	41.3	39.5
	10%	DB	39.3	39.9	42.9	48.6	56.9	61.2	62.8	62.8	56.8	49.7	43.2	40.8
		MCWB	37.2	36.6	38.2	41.8	47.7	53.0	56.2	56.1	51.7	46.2	40.0	37.9
	0.4%	WB	44.9	42.3	45.3	50.1	54.9	60.4	61.3	61.8	56.9	51.3	46.4	45.4
		MCDB	47.1	45.5	49.1	61.9	66.3	73.3	70.2	72.5	62.7	53.9	49.2	47.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.1	40.8	42.1	46.3	52.6	57.5	59.3	59.7	55.1	49.7	44.2	42.4
		MCDB	44.2	44.3	45.9	53.9	62.2	66.6	65.6	66.7	59.9	52.0	46.1	45.5
	5%	WB	39.4	39.3	40.4	44.3	50.7	55.4	58.0	58.1	53.7	48.0	42.1	40.2
		MCDB	42.2	42.8	44.0	50.3	58.5	62.2	63.6	63.7	57.4	50.4	44.3	43.1
	10%	WB	37.8	37.6	38.8	42.7	49.2	53.7	56.7	56.9	52.6	46.3	40.5	38.3
		MCDB	39.6	39.9	42.1	48.1	56.1	60.0	61.7	61.7	56.0	48.4	43.1	40.5
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	12.5	13.3	16.0	16.3	13.6	11.2	12.1	12.3	10.7	9.9	9.3
		MCDBR	9.3	11.8	15.6	25.4	26.2	22.0	17.5	21.4	20.7	13.7	9.3	9.8
		MCWBR	6.8	8.4	10.9	16.0	15.3	12.4	10.5	12.6	13.3	9.5	6.6	7.2
		MCDBR	8.9	10.3	13.7	20.2	21.4	18.4	15.4	16.7	13.9	10.3	8.3	9.8
	5% WB	MCWBR	6.9	7.7	10.0	13.3	13.0	10.8	9.5	10.4	9.8	7.7	6.4	7.6
(40) Clear-Sky Solar Irradiance	taub		0.287	0.293	0.317	0.369	0.390	0.388	0.389	0.361	0.340	0.324	0.285	0.256
	taud		2.144	2.326	2.297	2.203	2.254	2.349	2.392	2.499	2.537	2.435	2.217	2.110
	Ebn at Noon		167	236	263	264	266	269	266	267	256	222	168	139
	Edh at Noon		16	22	29	38	39	36	34	28	23	19	15	11
(44) All-Sky Solar Radiation	RadAvg		123	357	748	1236	1587	1606	1383	1113	710	374	165	83
	RadStd		15	62	95	135	169	115	141	138	114	57	25	10

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	+2.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page